

LABORATOIRE DE CHIMIE PHYSIQUE -
MATIÈRE ET RAYONNEMENT

Explainable Artificial Intelligence (XAI)

25 Juin 2025



Outline

01	Introduction	05	Discussion
02	Objectives	06	Conclusion
03	Methodology	07	Perspective
04	Results		

How We Study Pigments in Artworks:

- Use of MA-XRF, RIS, and LIS techniques combined in a single scanner.
- Non-invasive, multi-spectral approach allows analysis of pigments, varnishes, and binders.
- Enables detection of original layers, restorations, and even hidden compositions.

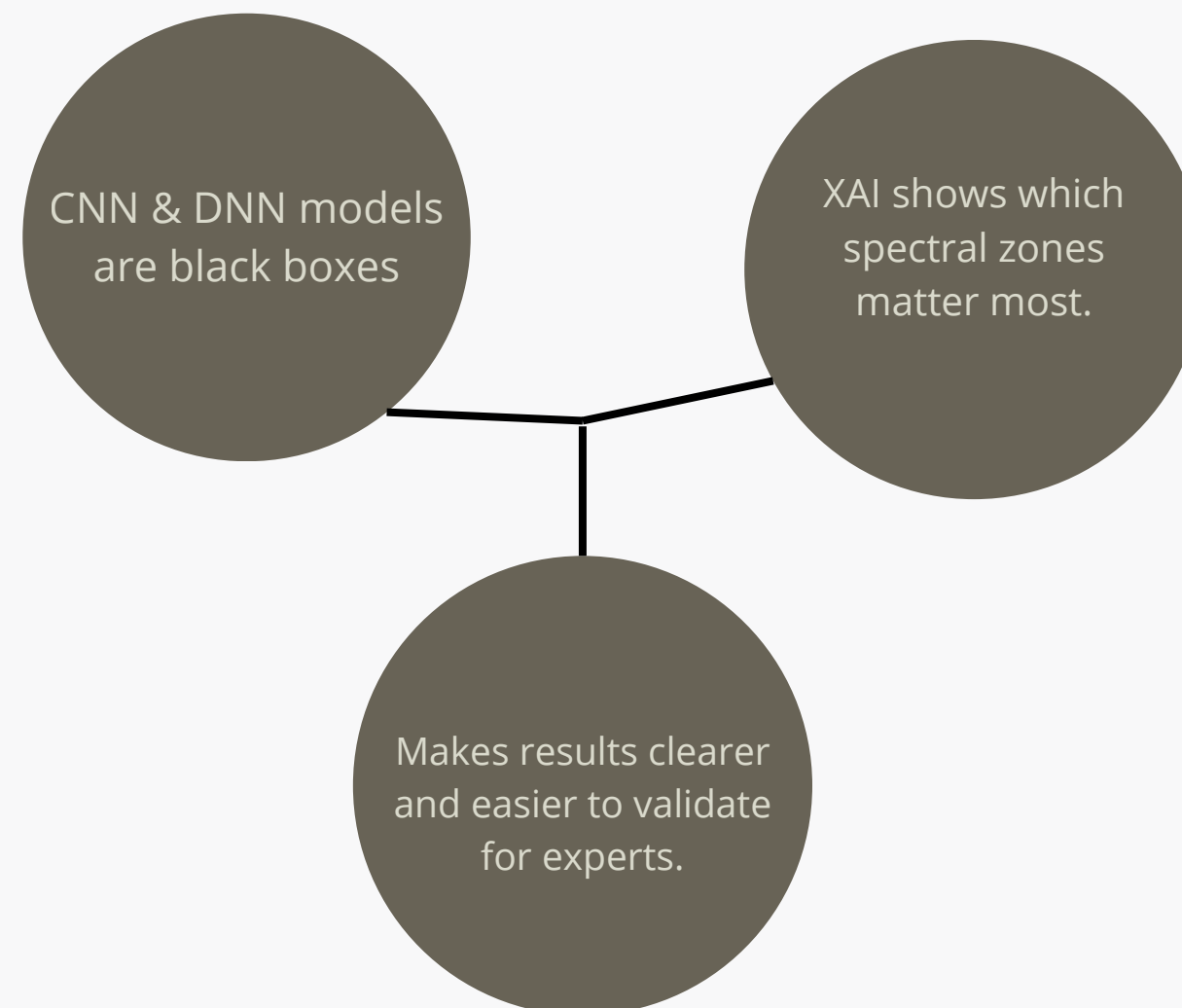
DATASET



Objective :

Use explainable AI methods to understand how a trained CNN or DNN model classifies pigments from hyperspectral data

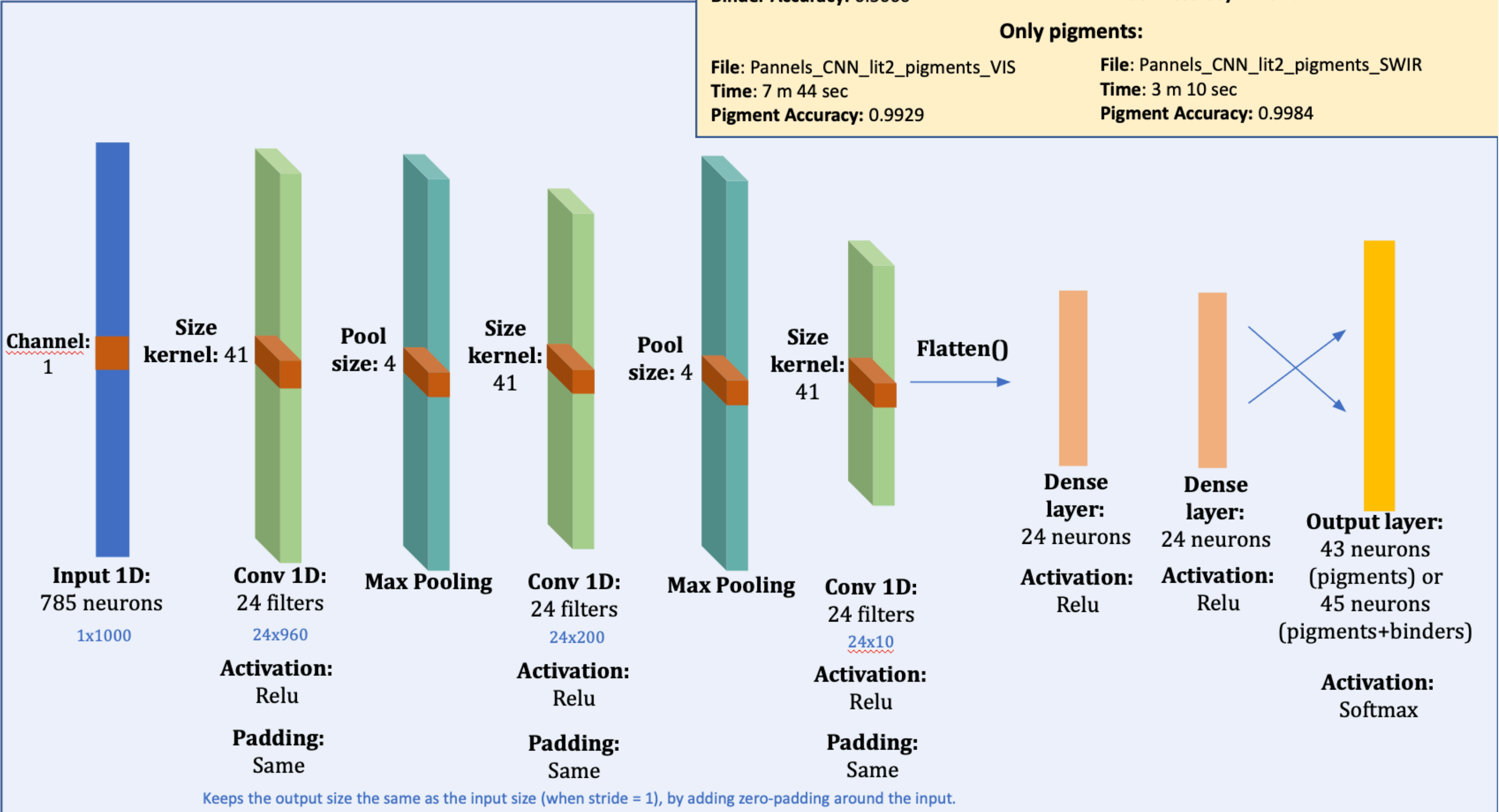
Why Explainability?



Convolutional Neural Network

Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., & Inman, D. J. (2021). 1D convolutional neural networks and applications: A survey. Mechanical Systems and Signal Processing, 151, 107398.
<https://doi.org/10.1016/j.ymssp.2020.107398>

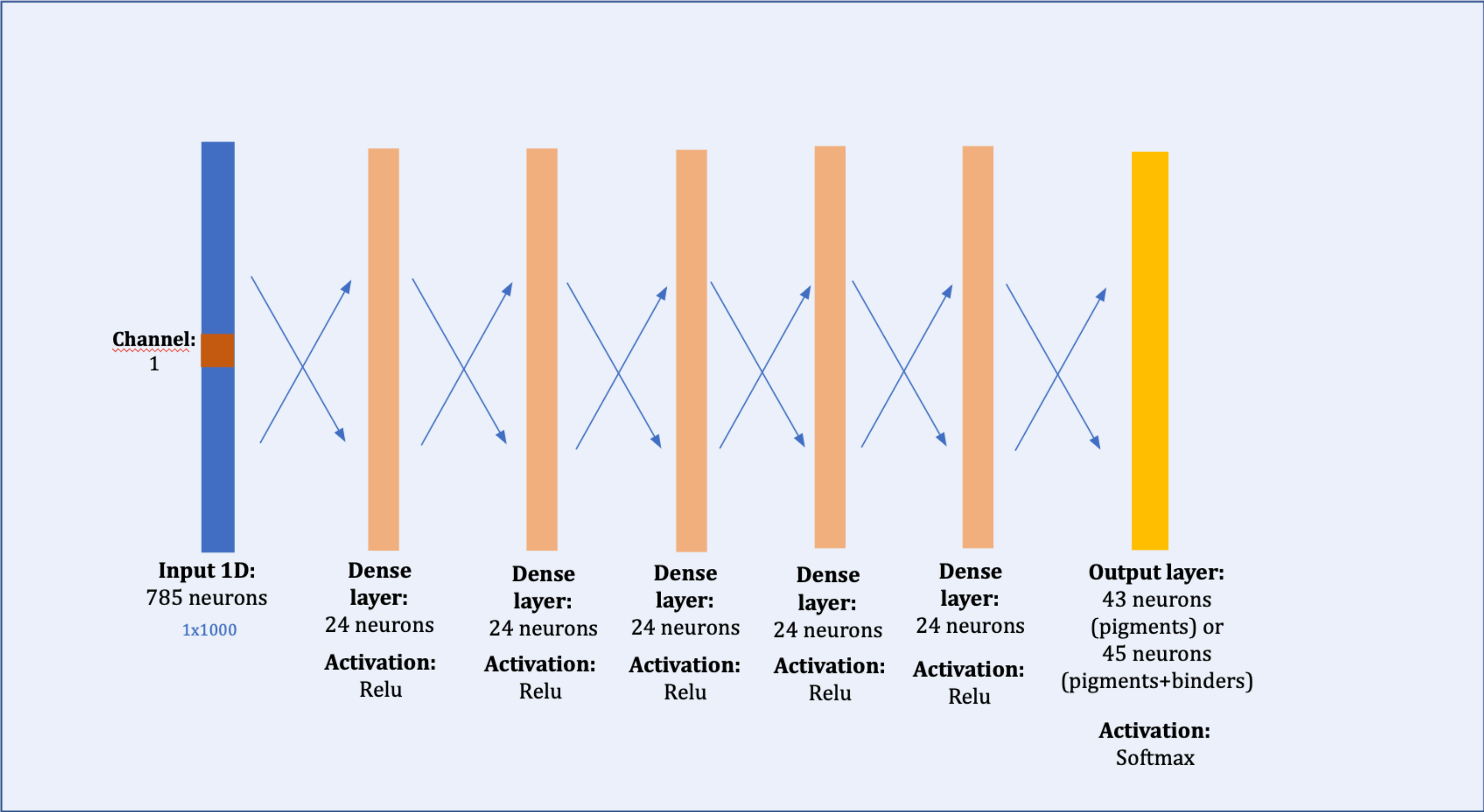
Epochs: 30 epochs for training
Batch size: 50
Sequence length: 1
Threshold: 0.5
Early stopping: patience: 10, restores best weights: true
Optimizer: Adam
Loss: Categorical_crossentropy
Metrics: Accuracy



Deep Neural Network

Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., & Inman, D. J. (2021). 1D convolutional neural networks and applications: A survey. Mechanical Systems and Signal Processing, 151, 107398. <https://doi.org/10.1016/j.ymssp.2020.107398>

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Steps:

1. Input Data:

Multimodal hyperspectral datasets

2. Spectral Zones Definition:

Divided full spectra into meaningful wavelength intervals (zones) based on:

- Known pigment absorption/emission features

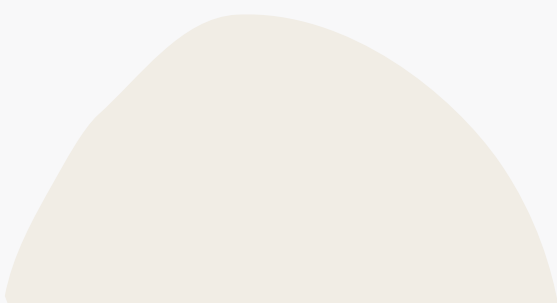
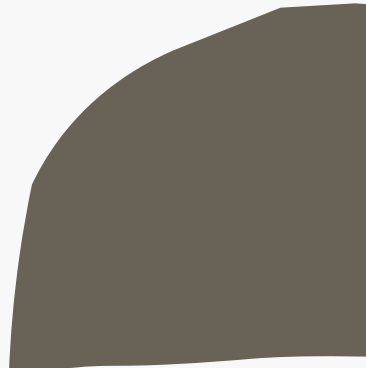
3. Explainability Techniques:

SHAP (SHapley Additive exPlanations)

LIME (Local Interpretable Model-agnostic Explanations)

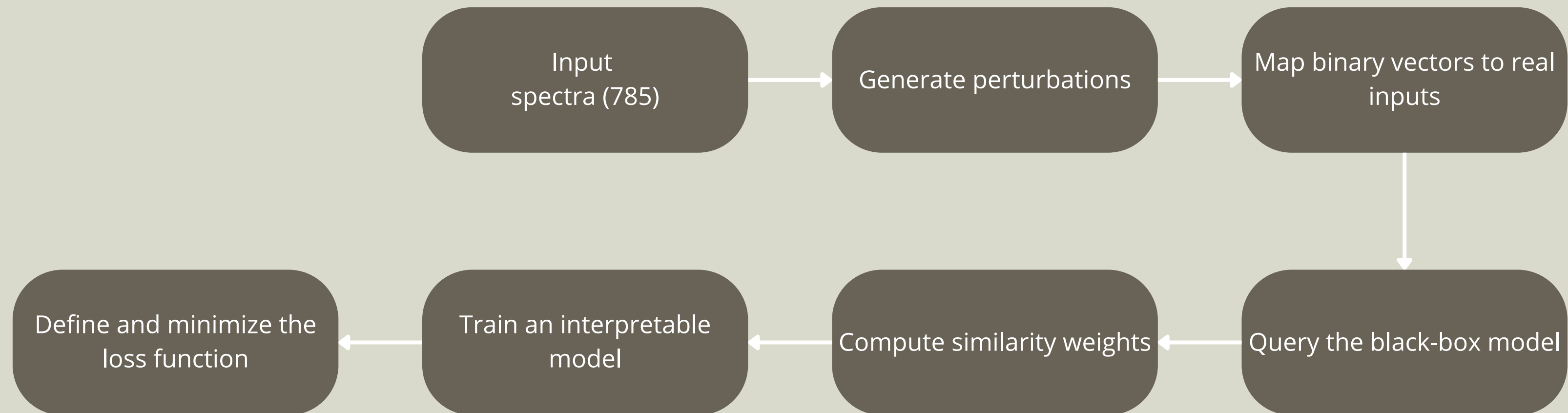
4. Output:

Bar plots showing which spectral zones had the greatest impact on CNN decisions.

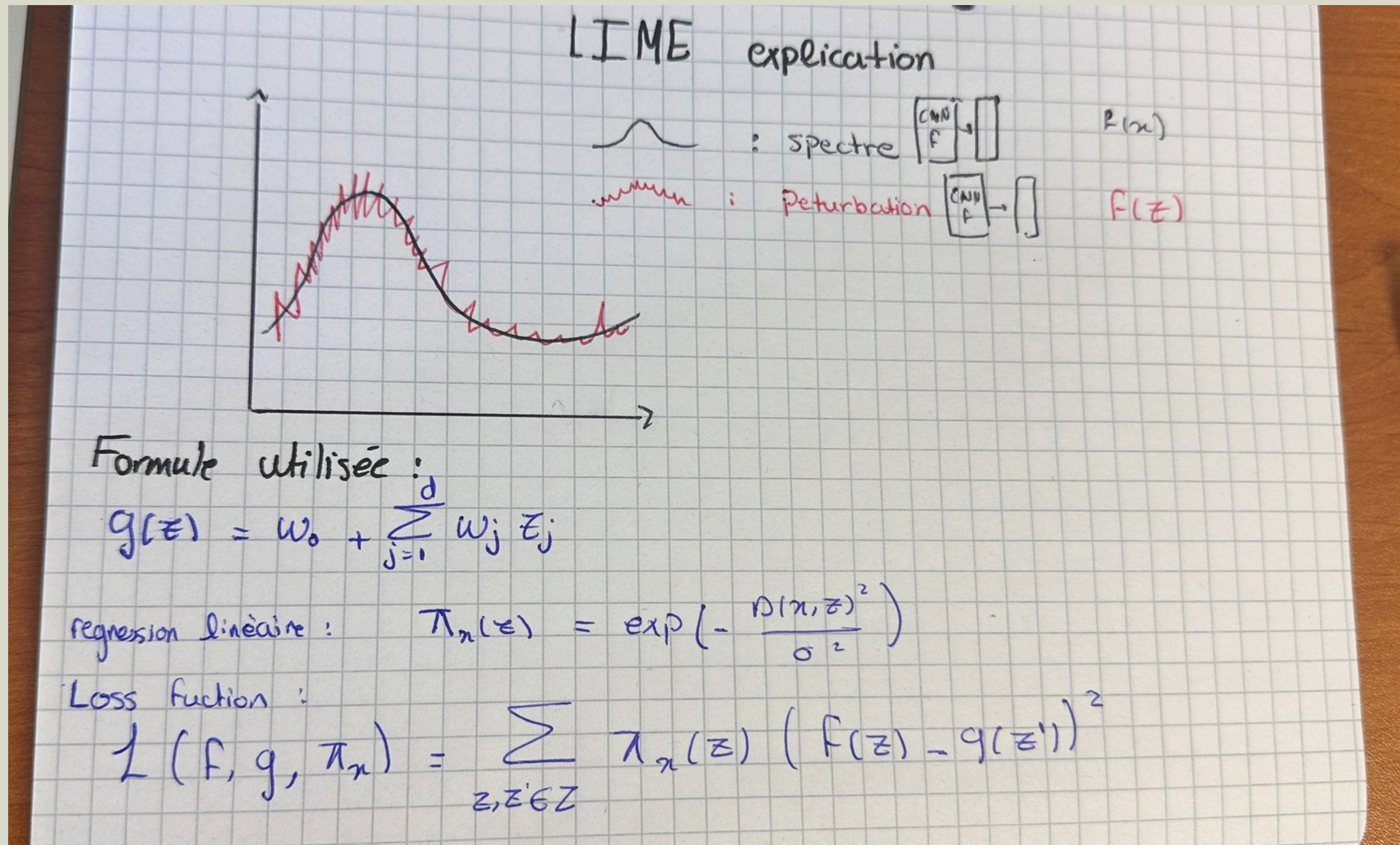


LIME

Local Interpretable Model-Agnostic Explanation



LIME



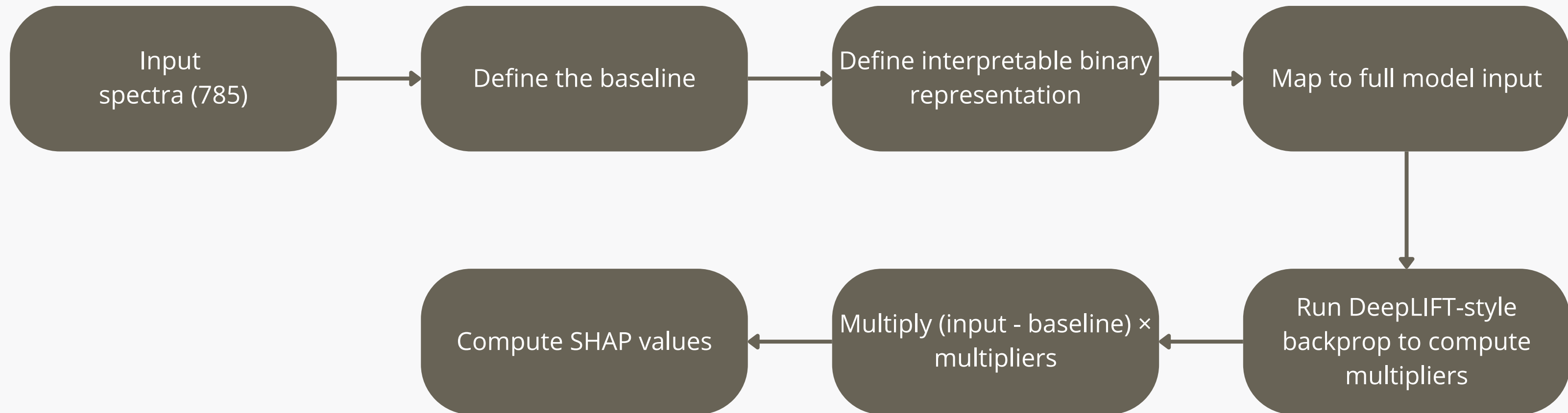
SHAP

SHapley Additive exPlanations

General Principle of SHAP

- It is based on Shapley values from game theory
- The goal: "How much does each feature contribute to the prediction?"

SHAP Explanation Pipeline (Deep SHAP)



DeepSHAP

problem:

I have 785 features (spectral bands) → impossible to evaluate 2^{785} subsets

Solution:

Use DeepSHAP

Deep SHAP Mathematical Approximation:

$$\phi_j \approx m_j \cdot (x_j - \mathbb{E}[x_j])$$

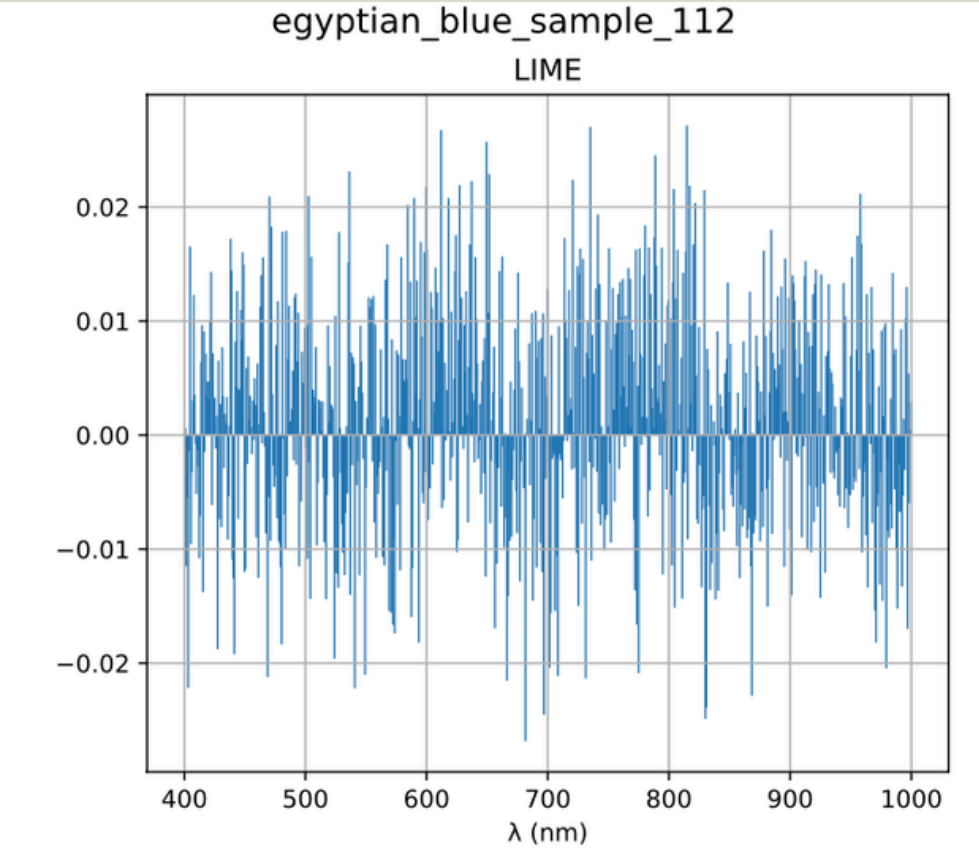
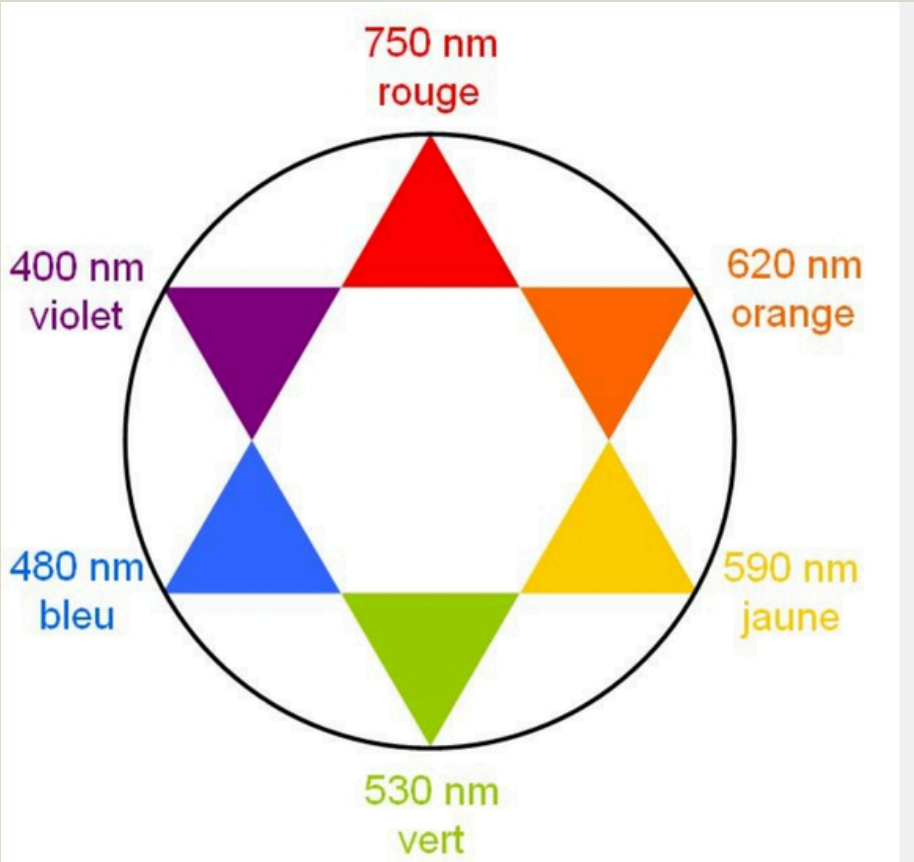
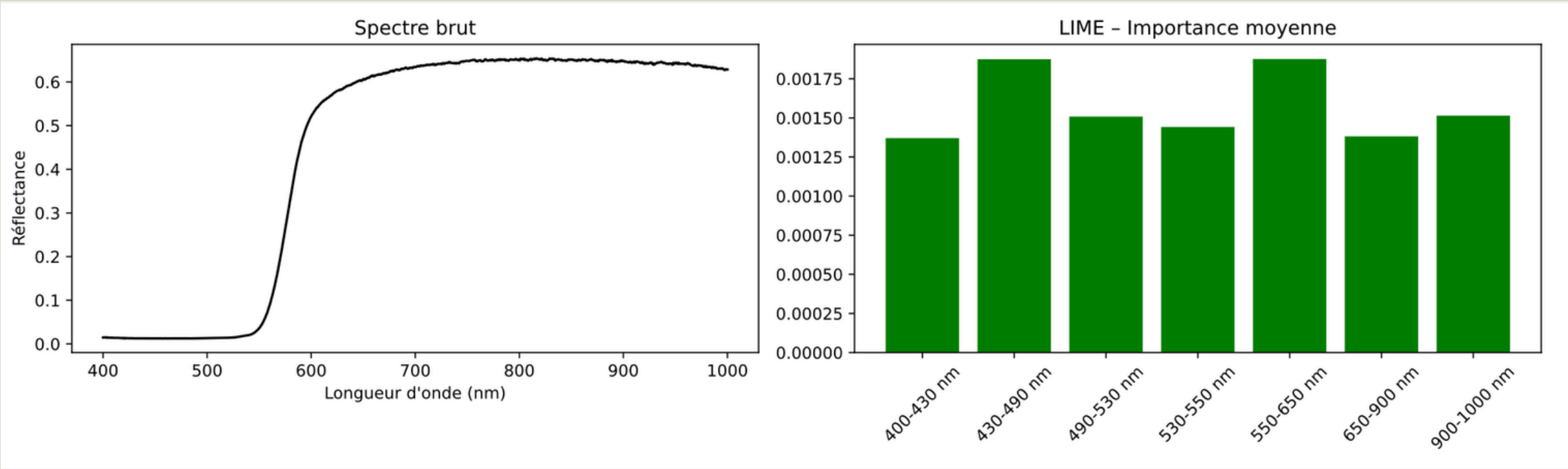
Where:

x_j : actual input value for feature j

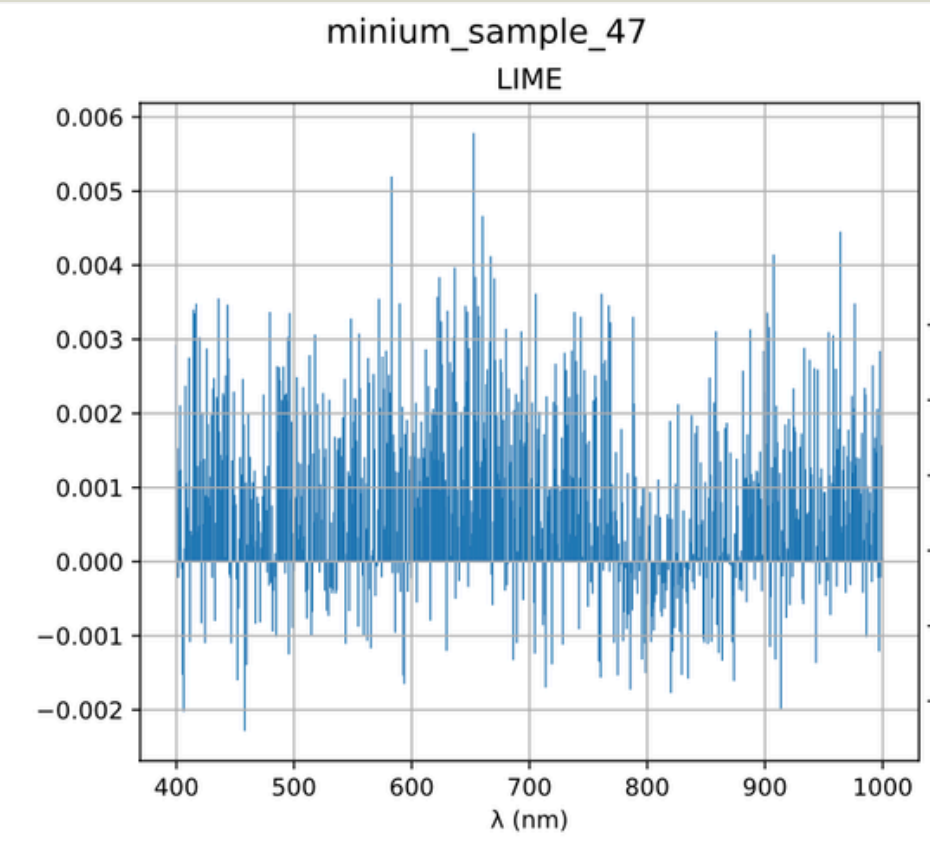
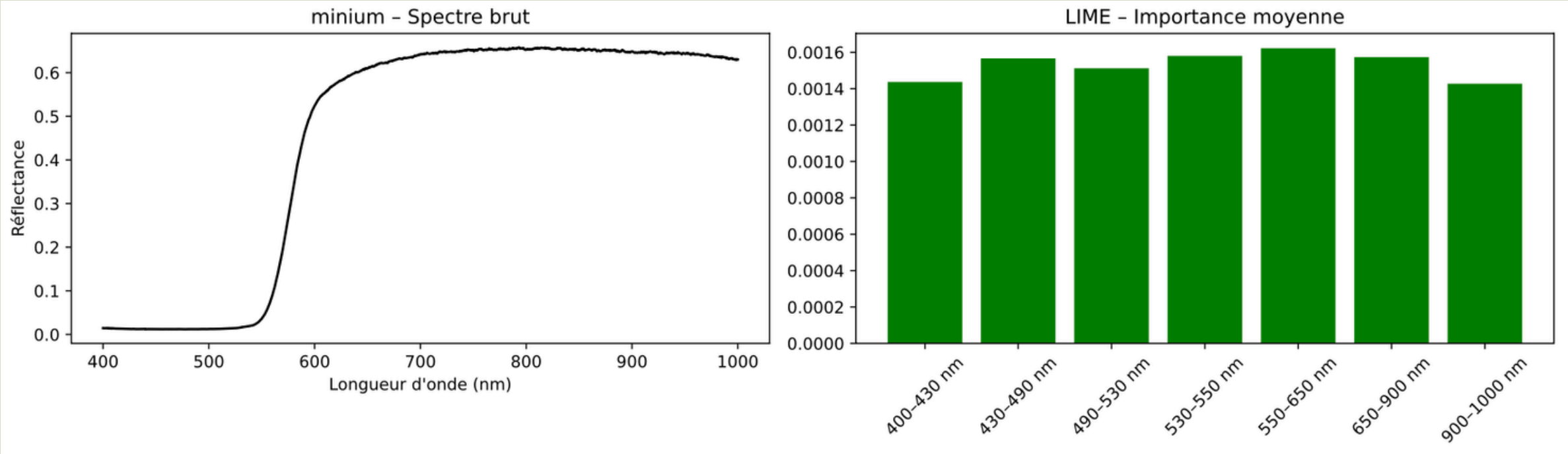
$\mathbb{E}[x_j]$: expected value from the background dataset

m_j : a value showing how much the model output changes when that feature changes.

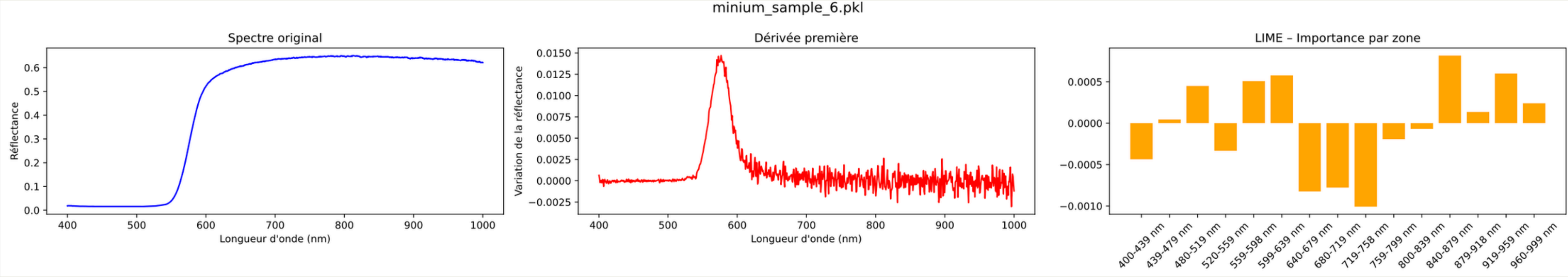
Results CNN VIS



Results DNN VIS



First derivative-CNN model -VIS



Conclusion	Limitations	Future Work
Spectral zones improve interpretability	Fixed spectral zone	Redefine zones
No clear "better" explanation method		Smoothing the spectra before computing derivatives



Any question?

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Thank you!!!

SAAB Alyaa

25/06/2025

Theoretical Foundation of SHAP (Shapley Values)

What is Shapley Value ?

Originally from cooperative game theory (by Lloyd Shapley, 1953):

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(M - |S| - 1)!}{M!} \cdot [f(S \cup \{j\}) - f(S)]$$

Where:

M: number of features (in your case, 785 spectral bands)

S: any subset of the features excluding feature j

f(S): the model output when only features in S are known

ϕ_j : the SHAP value for feature j, interpreted as its average marginal contribution

This is computationally intractable for large M, since it requires evaluating all 2^M subsets.

$$f(S \cup \{j\}) - f(S)$$

the marginal contribution of feature j to the prediction, in the context of subset S.

$$\frac{|S|! \cdot (M - |S| - 1)!}{M!}$$

the probability that subset S appears before feature j in a random permutation of all features.

KernelSHAP

Alternative Approach

Also uses the SHAP framework, but approximates

$$f(x) = \phi_0 + \sum_j \phi_j z'_j$$

$$\min_g \sum_{z'} \pi_{x'}(z') (f(h_x(z')) - g(z'))^2$$

$$\pi_{x'}(z') = \frac{(M-1)}{\binom{M}{|z'|} \cdot |z'| \cdot (M - |z'|)}$$

DeepLIFT

DeepLIFT computes how much each feature changed the output compared to a reference, not how sensitive the output is to small changes.

DeepLIFT guarantees that the sum of all feature contributions equals the total change in prediction between the input and the baseline.

Why it matters for Deep SHAP?

Deep SHAP uses DeepLIFT's rules to propagate importance values through each layer of the network, instead of sampling all feature combinations like Kernel SHAP.

This makes Deep SHAP:

- Much faster for large inputs (like 785 spectral bands),
- Still consistent with Shapley value theory,
- Applicable to deep models like CNNs.

$$\sum_{i=1}^n C_{\Delta x_i} \Delta o = \Delta o$$

$$\Delta x_j = x_j - r_j, \quad \Delta o = f(x) - f(r)$$

$$f(x) = f(\text{reference}) + \sum \phi_j$$

DeepLIFT

DeepLIFT (Deep Learning Important FeaTures) is a method that explains predictions by comparing a neuron's activation to a reference activation (usually the expected value of inputs).

It propagates contribution scores layer by layer, not partial derivatives.

Deep SHAP uses: DeepLIFT Multipliers

$$m_j = \frac{\Delta f}{\Delta x_j}$$

The SHAP value is approximated as:

$$\phi_j \approx m_j \cdot (x_j - \mathbb{E}[x_j])$$

Three Desirable Properties of SHAP

SHAP is the only method that satisfy these properties

1- Local Accuracy (Additivity)

$$f(x) = \phi_0 + \sum_{j=1}^M \phi_j$$

2-Missingness

$$x'_j = 0 \quad \Rightarrow \quad \phi_j = 0$$

3-Consistency

$$\forall z' \in \{0, 1\}^M, \quad f'_x(z') - f'_x(z' \setminus \{i\}) \geq f_x(z') - f_x(z' \setminus \{i\}) \quad \Rightarrow \quad \phi_i(f', x) \geq \phi_i(f, x)$$

for any context z' where feature i could be added,
If the model f' gives more credit to feature i than model f ,
Then the SHAP value ϕ_i must reflect that increase.

DeepSHAP

Propagation Rule (Chain of Multipliers)

Each operation in the network contributes multiplicatively

$$m_{y_i f} = \sum_j m_{y_i f_j} \cdot m_{x_j f_j}$$

- $m_{y_i f}$: Contribution totale de l'entrée x_i à la sortie y_i (ex: classe "pigment bleu").
- j : Un neurone dans une couche cachée
- $m_{y_i f_j}$: Contribution du neurone caché f_j à la sortie y_i .
- $m_{x_j f_j}$: Contribution de l'entrée x_j au neurone f_j .

And SHAP values are built layer by layer:

$$\phi_j = m_j \cdot (x_j - \mathbb{E}[x_j])$$

Background Reference $E[x]$

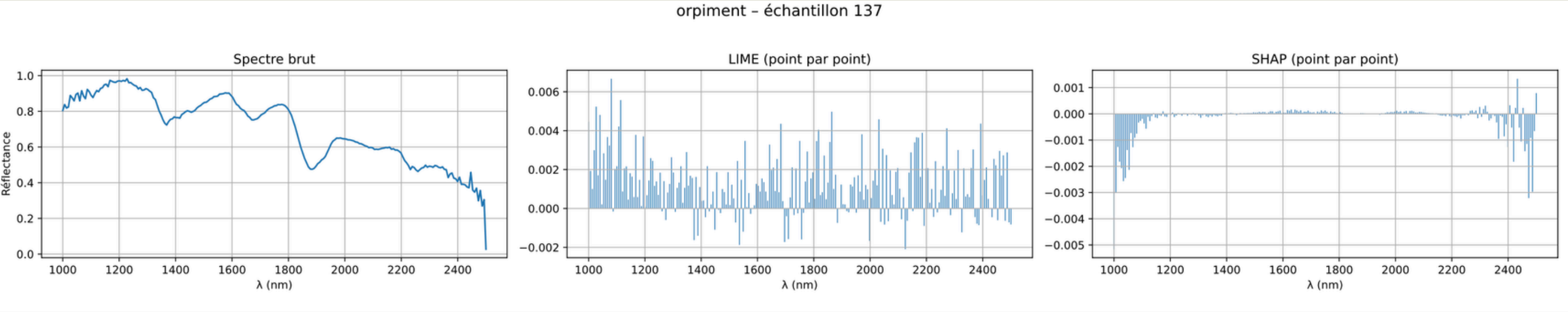
- Background of 50 representative spectra from the training set
- It is used to compute expected values $E[x_j]$
- Replacing a feature with its expected value simulates it being "unknown"

This ensures SHAP values satisfy:

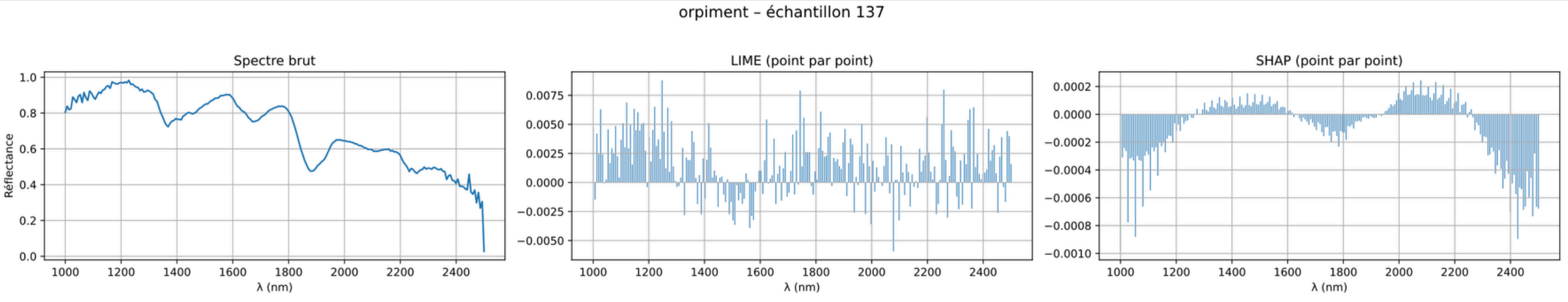
- Missingness
- Local accuracy

Results CNN-DNN SWIR

SHAP/LIME without
zones-DNN



SHAP/LIME without
zones-CNN



Definy the wavelength

